

ENGR. ABU KOWSAR, PMP®, PMI-ACP®

Programmer | Department of ICT, Government of Bangladesh

Technical Project Manager | AI & Digital Transformation Researcher

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ABOUT ME

Results-oriented ICT professional and Technical Project Manager with 12+ years of experience spanning government digital transformation, software engineering, AI-enabled solutions, project management, cybersecurity, data analytics, and capacity development. Currently serving as a Programmer with the Department of ICT, Government of Bangladesh, with field-level responsibility for translating national digital priorities into practical, scalable solutions. PMP® and PMI-ACP® certified, with a strong record of coordinating multidisciplinary initiatives, developing digital platforms, designing training programs, and applying Agile and data-driven approaches to improve delivery, service quality, and citizen outcomes.

CORE PROFESSIONAL COMPETENCIES

- Technical Project & Program Management | PMP® | PMI-ACP® | Agile/Scrum | PMO & Delivery Governance
- AI & Digital Transformation | Generative AI | AI-assisted productivity | Human-centered AI | Bangla AI/LLM concepts
- Government Digital Services | e-Governance | Smart Government | District-level ICT Operations | Digital Ecosystems
- Software Engineering | Next.js | React | Node.js/Express | REST APIs | MongoDB/Mongoose | JavaScript | Git/GitHub
- Cybersecurity & Resilience | SOC concepts | SIEM | NIST Incident Response Lifecycle | OWASP Top 10
- Data & Analytics | SQL | Oracle | MongoDB | Data visualization | KPI frameworks | Dashboard design
- Stakeholder & Team Leadership | Cross-functional coordination | Training & facilitation | Change management
- Technical Documentation | Concept Notes | DPP/PC-1-style proposals | SOPs | Gantt/implementation plans | Training manuals

PROFESSIONAL EXPERIENCE

Programmer

Department of ICT, ICT Division, Ministry of Posts Telecommunications and IT | 25 Aug 2019 - Present | Bangladesh

- Lead and coordinate district-level ICT initiatives, digital service delivery, technical support, and implementation activities aligned with national digital-transformation priorities.
- Translate institutional and field-level requirements into project concepts, implementation plans, technical specifications, dashboards, SOPs, and operational workflows.
- Coordinate with government stakeholders, local institutions, service providers, trainers, and technical teams to deliver technology-enabled programs.
- Support digital skills, freelancing, AI, cybersecurity, smart-government, and citizen-service initiatives through planning, execution, monitoring, and reporting.

Key impact: Recognized for combining public-sector delivery discipline with hands-on software, AI, project-management, and capacity-development expertise.

Assistant Programmer

Palli Daridro Bimochon Foundation (PDBF), Ministry of Local Government | 2015 - 2019 | Bangladesh

- Supported ICT operations, software applications, data management, technical troubleshooting, and digital workflow improvement.
- Participated in technology-enabled organizational initiatives and supported users across operational functions.
- Applied programming, database, systems-analysis, and ICT support skills to improve reliability and service delivery.

Technical Project Manager

Bangladesh Canada Friendship Tech Ltd. | 2013 - 2015 | Bangladesh

- Designed and prototyped digital platforms using modern web technologies, APIs, databases, GIS, and cloud deployment patterns.
- Developed training curricula and professional manuals covering AI productivity, project management, cybersecurity, programming, and digital skills.

SELECTED STRATEGIC PROJECTS & INITIATIVES

- One Student One Skill (OSOS) - skills-development platform concept focused on measurable student skills, employment, and entrepreneurship outcomes.
- National STEM & AI Platform / School of Robotics - digital learning, STEM education, AI literacy, and robotics ecosystem concepts.
- Bangladesh AI Citizen Assistant - concept for AI-enabled citizen support and accessible public-service information.
- District AI Innovation Hub - proposed district-level innovation, training, startup, and applied-AI support ecosystem.
- District ICT Operations Dashboard - concept for consolidated monitoring of ICT assets, services, projects, and operational KPIs.
- District Cyber Safety & Incident Response Network - proposed local cyber-awareness and incident-response coordination model.
- National ICT Asset & Lab Management System - digital inventory, lifecycle, maintenance, and monitoring concept for ICT assets.
- Smart Government / Citizen Service Platforms - design and planning work across e-governance, digital employment, service delivery, and data-driven administration.

EDUCATION & ACADEMIC TRAINING

Master of Science (M.Sc.) in Computer Science & Engineering

Islamic University, Bangladesh | 2009 | Bangladesh

Advanced study in computer science and engineering with emphasis on software, systems, computing, and applied technology.

Bachelor of Science (B.Sc.) in Computer Science & Engineering

Islamic University, Bangladesh | 2008 | Bangladesh

Foundation in computer science, software engineering, algorithms, databases, computer networks, and systems.

Postgraduate Diploma (PGD) in ICT

Bangladesh | 2015

Postgraduate professional development in information and communication technology.

PROFESSIONAL DEVELOPMENT & CERTIFICATIONS

- PMP® - Project Management Professional
- PMI-ACP® - Agile Certified Practitioner
- Agile & Scrum - professional training and applied project-delivery practices
- Lean Six Sigma White Belt - 2025
- IBM AI Developer Professional Certificate
- Oracle Certified / Oracle Database-related professional credentials
- Additional professional development in AI engineering, cybersecurity, software development, data analytics, and digital transformation

TECHNICAL SKILLS & TECHNOLOGY STACK

- Programming & Web: JavaScript, Python, React, Next.js, Node.js, Express, REST API architecture, Axios
- Backend & Data: MongoDB, Mongoose, SQL, Oracle, database design, API integration, authentication concepts
- Cloud & Deployment: Vercel, Render, Azure concepts, cloud-based application deployment and operations
- AI & Data: Generative AI, prompt engineering, AI-assisted software development, machine learning concepts, data analytics
- Cybersecurity: SOC/SIEM concepts, NIST Incident Response Lifecycle, OWASP Top 10, cyber-awareness and resilience
- Project & Productivity: MS Office, documentation, requirements analysis, Agile/Scrum, project planning, KPI dashboards
- Design & Collaboration: Figma, Git, GitHub, GIS concepts, OpenStreetMap, technical prototyping

LEADERSHIP, MANAGEMENT & CAPACITY DEVELOPMENT

- 12+ years of progressive experience across ICT implementation, project delivery, technical coordination, and organizational capacity development.
- Lead cross-functional activities involving government officials, trainers, developers, vendors, educational institutions, and community stakeholders.
- Develop and facilitate professional training programs in AI productivity, project management, cybersecurity, programming, and digital skills.
- Prepare concept papers, project proposals, implementation roadmaps, SOPs, monitoring frameworks, Gantt charts, budgets, and training schedules.
- Mentor learners and professionals in programming, AI, project management, digital transformation, and career-oriented technology skills.

SELECTED TRAINING & KNOWLEDGE AREAS

- AI for Project Management | Generative AI | Prompt Engineering | AI Productivity | AI Content Creation
- PMP/PMBOK-based project management | Agile | Scrum | Risk | Stakeholder | Scope | Schedule | Cost | Quality
- Cyber Safety | SOC Analyst fundamentals | SIEM | NIST Incident Response | OWASP Top 10
- Full-stack Web Development | Next.js/React | Express APIs | MongoDB/Mongoose | Authentication | Cloud Deployment
- Data Analytics | SQL | Dashboard/KPI design | Digital government data and service monitoring

PROFESSIONAL ACHIEVEMENTS & IMPACT

- Designed and advanced multiple government-focused digital transformation concepts linking technology investment to measurable citizen, employment, skills, and service-delivery outcomes.
- Developed structured implementation frameworks for district-level digital initiatives, including KPIs, work plans, training models, monitoring mechanisms, and operational governance.
- Led the conceptualization of skills and employment initiatives such as One Student One Skill, with outcome-oriented targets for skills acquisition, employment, and entrepreneurship.
- Created professional training manuals and curricula for AI, project management, cybersecurity, programming, and digital productivity.
- Combined project-management certifications with practical engineering experience to improve planning, coordination, documentation, and delivery discipline.

RESEARCH, PUBLICATIONS & KNOWLEDGE CONTRIBUTION

- Technical and professional writing on AI engineering, AI-powered project management, cybersecurity, digital transformation, and ICT-enabled public services.
- Concept development and research-oriented work on Bangla AI, AI citizen assistance, STEM/robotics education, digital employment, cyber safety, and smart-government systems.
- Contributor to training and knowledge resources designed to translate emerging technologies into practical skills for professionals, students, and public-sector stakeholders.

AWARDS, RECOGNITION & PROFESSIONAL STANDING

- Project Management Professional (PMP®) - internationally recognized project-management credential.
- PMI Agile Certified Practitioner (PMI-ACP®) - internationally recognized Agile credential.
- Lean Six Sigma White Belt - process-improvement credential.
- IBM AI Developer Professional Certificate - applied AI/software development credential.
- Recognized professional focus in government digital transformation, AI-enabled innovation, ICT capacity development, and technology project delivery.

PROFESSIONAL MEMBERSHIPS & AFFILIATIONS

- Project Management Institute (PMI) - professional affiliation through PMP® / PMI-ACP® credentials.
- Professional engagement with ICT, software engineering, AI, cybersecurity, project management, and digital-transformation communities.
- Collaboration with government, education, training, and technology stakeholders on digital initiatives.